

SYLLABUS CONTENT

By the end of this chapter, you should be able to:

- ▶ A4.4.1 Discuss the ethical implications of machine learning in real-world scenarios
- ▶ A4.4.2 Discuss ethics as technologies become integrated into daily life

A4.4.1 Ethical implications

**Does all knowledge impose ethical obligations on those who know it?**

Discuss the ethical use of machine learning, especially in sensitive areas like surveillance or decision-making.

In surveillance (like facial recognition), concerns about privacy, consent and surveillance biases abound. Surveillance systems can be used to monitor, control and sometimes discriminate against populations.

With the involvement of machine learning in decision-making, such as in hiring, lending and law enforcement, these systems can influence people's lives significantly, and have been shown to inherit and amplify biases present in the training data.

Machine learning models – inherently knowledge-driven systems – are based on data that encapsulate various forms of knowledge, from human behaviour to biological patterns. As creators and users of these systems, there is a responsibility to ensure that this knowledge is used ethically.

The world is changing rapidly. Advances in technology, including those in machine learning, pose significant challenges and questions for us as a society. It is important to take some time to weigh these ethical questions and not get caught up by the shiny new tech without taking the time to think through how it will impact us and those around us. The following are some of the ethical issues to consider.

- **Accountability:** With whom or where does responsibility lie for decisions made by machine learning systems? Is it with the company that produced the AI or the people using it? Is it a blend of both? Is it possible to determine how and why a machine learning system made a particular decision?

One incident that highlights the issue of accountability involved a self-driving car. The driver who was behind the wheel of a self-driving car when it hit and killed a pedestrian in 2018 pleaded guilty to endangerment and was sentenced to three years of supervised probation.

- **Algorithmic fairness and bias:** Machine learning can perpetuate existing social bias if it is present in the training data, or if the model's design knowingly or unknowingly favours certain groups. Fairness requires actively identifying and mitigating bias in the data set and algorithms.

COMPAS is a recidivism algorithm used by many US court systems. It has been found to have racial bias, predicting higher risk of recidivism for black people and lower risk of recidivism for white people.

Another example is that, in 2018, Amazon scrapped a “secret” AI recruiting tool that was biased against women.

Finally, generative AIs have a constant challenge regarding reinforcing and exacerbating stereotypes and bias.

- **Consent:** Large data sets used for training regularly contain information collected without explicit consent. Many large companies are now performing machine learning on their customer databases, or selling their customer data to other data-matching companies. How much control should people retain over their personal information?

Google’s DeepMind was found to be in breach of UK privacy laws after it failed to adequately inform patients about the use of their personally identifiable health data in developing an app to detect kidney injuries.

- **Environmental impact:** Machine learning models require enormous computational power, especially in the training phase. This leads to substantial energy consumption and implications for carbon emissions.

Cornell University scientists found that training LLMs (large language models) like GPT-3 consumed an amount of electricity equivalent to 500 metric tons of carbon. In fact, DatacenterDynamics reports global power use by data centres will more than double from 460 TWh in 2022 to over 1000 TWh in 2026.

- **Privacy:** Machine learning systems can predict or classify personal behaviour in ways that invade personal privacy. The capacity of machine learning systems to apply inference means privacy may be further compromised by systems deducing health conditions not even provided to the model.

In 2018, fitness tracking app Strava released a global heat map of user activities that inadvertently revealed the locations of secret military bases and patrol routes, showcasing a significant privacy leak.

- **Security:** Machine learning systems can be vulnerable to attack through a variety of means. Three common attacks include:
 - **data poisoning**, which involves introducing untrue or harmful data into the training data set to manipulate the model for nefarious purposes
 - **model evasion**, where input (such as prompts) is used to “trick” the model into making incorrect outputs against its training (sometimes known as “jailbreaking”, in the context of generative AI)
 - **model inversion**, referring to gaining access to sensitive data contained within the training data.

Within 24 hours of release, Microsoft’s Tay Twitter Bot was manipulated through malicious input data to produce grossly inappropriate and offensive tweets.

When GPT-3 was first released by OpenAI, it lacked many of the filters now present and it was trivially easy to engineer prompts that produced foul, toxic or illegal content.

- **Societal impact:** Machine learning is increasingly disrupting employment markets, and influencing public opinion. There is a careful balance between technological advancement and maintaining social welfare, which needs to be considered.

Clearview AI, which scrapes billions of photos from the internet for facial recognition, has raised societal concerns about surveillance, consent and civil liberties.

- **Transparency:** Most engineers cannot explain how their systems generate the outputs they create, especially those that use neural networks. The best that can be done is to

point to the training data rather than the algorithm itself. This lack of transparency, or human understandability, of what these algorithms are and how they work poses significant questions.

In 2019, tech entrepreneur David Heinmeier Hansson wrote on X (formerly Twitter) that Apple Card offered him 20 times the credit limit of his wife, although they have shared assets and she has a higher credit score, raising questions about the transparency of the algorithms used for financial decision-making.

- **Bias in training data:** Bias in training data is a core challenge for machine learning. Over- or under-representation of particular demographics will affect the model's predictions and reliability. Rigorous data collection, processing and evaluation methods are required to ensure broad and fair representation.
- **Misinformation:** Machine learning can generate and spread false information with ease, making it very difficult to ensure accurate and reliable communication online. As generative AI, in particular, becomes increasingly realistic and convincing in its outputs, it will become almost impossible to avoid falling victim to fake news, fake images and fake videos.

It is believed that misinformation on Facebook received six times more clicks than factual news during the 2020 US election, according to a study by NYU.

As generative AI deep fakes become weapons of the political debate, confusion over what to believe will only pose more complex challenges in the future.

- **Bias in online communication:** Machine learning-based recommendation systems are designed to maximize user engagement on a platform. One method of doing this is by recommending more of the same kinds of content that users have previously engaged with. This can create “echo chambers” that reinforce existing beliefs and minimize alternative viewpoints.

Facebook newsfeed algorithms and YouTube's recommendation systems have both been criticized for creating filter bubbles and echo chambers, where users are predominantly shown content that aligns with their existing beliefs, potentially polarizing public opinion.
- **Online harassment:** Machine learning can be used to automate harassment on an enormous scale. Bots can troll and target individuals or groups with ease, and can increasingly make it seem like the attacks are coming from people. Generative AI is being used to create deep fakes in hurtful and abusive ways that authorities are struggling to keep up with.
- **Privacy and anonymity in online communications:** Users often are not aware of or do not fully understand how their data are used and processed by machine learning algorithms. Users may think their actions are anonymous, but increasingly machine learning algorithms can perform de-anonymization with a high degree of reliability. There is very little awareness of this in the broader community.

In 2006, Netflix released a data set containing 100 million movie ratings from 500,000 subscribers, intended for use in a global competition to improve the accuracy of Netflix's recommendation algorithm. The data was supposedly anonymized by removing any personal identifying information. Researchers from the University of Texas at Austin demonstrated that it was possible to re-identify users by comparing the anonymized Netflix data with publicly available movie ratings on the Internet Movie Database (IMDb). Using only a small amount of additional information about an individual's preferences, the researchers were able to identify personal viewing habits and potentially sensitive information.

A4.4.2 Reassessing ethics as technologies become further integrated

As artificial intelligence and other technology continues to advance and evolve over the years ahead, society is going to need to regularly reassess the implications from an ethical viewpoint. There are many challenges that lie ahead; the following list is just a discussion starter.

- **Quantum computing:** Quantum computing could potentially break many of the cryptographic systems that currently secure digital communications and cryptocurrencies. The development of quantum-resistant cryptography is an important area of research that needs to be prioritized.
- **Augmented reality:** AR can collect vast amounts of personal data about users' environments. Additionally, what are the ethics around altering a person's perception of reality? Does this disconnect them from the society of which they are part, resulting in a loss of empathy?
- **Virtual reality:** As VR becomes more realistic, what are the mental-health concerns for those who use the systems excessively or for escapism? What should the limits be when it comes to VR being used to access violent or explicit material?
- **Pervasive AI:** How do we guard against intrusive surveillance and the seemingly never-ending collection of our personal data for use in machine learning data sets?
- **Privacy:** Who owns the data about you? Is it you, or the company that collected it? As data collection becomes more complex, will there be a move towards more transparent and informed consent about what happens with our personal information?
- **Equity:** How can we ensure that advances in technology reduce rather than magnify equitable access to technology across socio-economic, racial, gender, social and geographical groups?

Top tip!

This section shared real-life case studies on the impact of many of the ethical questions being raised by this topic. Be familiar with case studies that you can refer to in your exam responses. If you can discuss with specificity a relevant situation that occurred, it goes a long way towards demonstrating that you care about the issue.

Common mistakes

Students make a number of common errors when addressing ethics-related questions, which extends to the discussion of machine learning.

- **Don't oversimplify the issues.** Avoid reducing complex ethical issues to simple right or wrong answers. The ethical implications of machine learning are nuanced and often involve interconnected considerations of accountability, fairness and societal impact.
- **Don't confuse technical bias with ethical bias.** Distinguish between technical bias (deviation in an algorithm that leads to less accurate predictions) and ethical / social bias (prejudices in data that lead to unfair outcomes for certain groups).
- **Don't limit your responses to issues of privacy and security.** Consider a broader range of ethical issues, such as environmental impact, societal changes and the implications for mental health. Show you have a deep understanding of the complexities involved, rather than taking the lazy approach of resorting to an exam response that discusses privacy or security superficially.
- **Don't neglect the importance of reassessment.** Ethics guidelines can never be static, as technology and its impact on society is not static.

ACTIVITY

Social skills: Set up a class debate or panel discussion where you argue the ethical implications of using machine learning, such as bias, privacy and transparency concerns. Facilitate peer feedback sessions where you review and provide constructive criticism on each other's machine learning projects or presentations.

Some possible debate prompts include:

- Should health-insurance companies have access to predictions about potential future illnesses to set premiums, even if this could lead to higher costs for those deemed at higher risk?
- Should autonomous vehicles be programmed to prioritize the lives of pedestrians over the life of the vehicle's passenger(s)? How should these ethical decisions be programmed into autonomous systems?
- Is it ethical to use a recruitment tool that shows bias towards certain educational institutions? Should the company stop using it until it can be proven to be unbiased?
- If a city implements widespread facial recognition through CCTV cameras to reduce crime, is this worth the lessening of privacy or the risk of false accusation?
- Should social-media platforms be held responsible for breaking echo chambers and ensuring a balanced exposure to different viewpoints? How can this be balanced with business models that require maximizing engagement to earn revenue?

REVIEW QUESTIONS

- 1 An AI company develops a facial recognition system used in public surveillance.
 - a Outline three ethical implications of using facial recognition technology in public spaces.
 - b
 - i Identify two potential biases that could arise in facial recognition systems.
 - ii Outline the societal impacts of each.
 - c Outline two measures that could be implemented to address these ethical concerns and biases.
- 2 A social-media company uses algorithms to personalize newsfeeds based on user interactions.
 - a Outline two potential ethical issues related to algorithmic bias in personalizing newsfeeds.
 - b Outline two strategies the company could implement to ensure the ethical use of personalization algorithms.
 - c
 - i Identify two implications of lack of transparency in algorithmic decision-making.
 - ii Outline two methods to improve transparency in algorithmic decision-making.
- 3 A university uses AI to make admissions decisions based on application materials.
 - a Outline three potential ethical concerns with using AI in university admissions.
 - b
 - i Outline two possible biases that could arise in this AI system.
 - ii Outline their impact on students.
 - c Describe measures to address these ethical concerns and biases.

● Linking questions

- 1 How can machine learning be applied to optimize network traffic management? (A2)
- 2 How does database programming in SQL differ from programming computationally in a high-level language? (A3, B2)
- 3 To what extent are developments in machine learning ethical? (TOK)
- 4 How can larger models be processed using GPUs and cloud processing? (A1)
- 5 Can machine learning find and improve network security problems? (A2)

EXAM PRACTICE QUESTIONS

1 Health monitoring app

A tech startup has developed a health monitoring app that uses machine learning to predict potential health issues based on user-inputted symptoms, lifestyle data and historical health data. The app classifies user health into categories such as "low risk", "medium risk" and "high risk".

- a i State whether this system should be classified as artificial intelligence or machine learning. [1]
- ii Outline one reason for your choice. [2]
- b Describe the potential need for specialized hardware (e.g. GPUs) in deploying this app on mobile devices. [2]
- c Describe the importance of data cleaning in this scenario, particularly addressing missing values in lifestyle data. [2]
- d Describe how feature selection could impact the accuracy and efficiency of the predictive model used in the app. [2]
- e Suggest the type of machine learning algorithm that would be suitable for this classification task. [4]
- f Outline the implications of choosing a high value of k in a k -nearest neighbours (KNN) algorithm for this application. [2]
- g Outline three ethical concerns related to privacy and data security in health-related apps. [3]
- h Describe two measures that could be implemented to address potential biases in the data set, especially relating to underrepresented groups. [2]

2 Autonomous public transport system

A city plans to implement an autonomous bus service that uses machine learning to optimize routes based on traffic patterns, weather conditions and passenger demand.

- a i Define "edge computing". [1]
- ii Describe its relevance in real-time data processing for autonomous vehicles. [2]
- b Describe whether a deep learning model would be more effective than a traditional machine learning model for processing complex environmental data. [2]
- c Describe how data normalization affects the performance of machine learning models dealing with varied data types such as weather conditions and traffic density. [2]
- d i Identify a common data quality issue that might arise with real-time traffic data. [1]
- ii Outline a preprocessing step to mitigate these issues. [2]
- e Describe how reinforcement learning could be applied to optimize bus routes dynamically. [2]
- f Discuss the potential use of transfer learning from other cities' traffic management systems to improve route optimization. [2]
- g Describe the ethical implications of using surveillance data (e.g. from traffic cameras) in training machine learning models for public transport systems. [2]
- h Describe the societal impacts of replacing human-driven buses with autonomous buses, including job displacement and public safety. [2]

3 AI-powered recruitment tool

A multinational corporation implements an AI-powered tool to screen job applications and predict the suitability of candidates based on their résumés and answers to pre-interview questions.

- a Describe the classification vs regression nature of the predictive model used by the AI tool. [2]
- b Describe the impact of processing speed and memory requirements on the scalability of the AI tool across the corporation's global offices. [2]
- c i Identify potential biases in the training data set. [1]
- ii Outline how these could be mitigated during data preprocessing. [2]
- d Describe the importance of feature selection in improving the predictive accuracy of the AI tool. [2]
- e Describe the use of a decision tree model over a regression-based model for this classification task. [3]

- f Outline two ethical concerns related to AI decision-making in recruitment, particularly in terms of fairness and transparency. [2]
 - g Describe two methods to ensure the ethical use of AI in recruitment with respect to increasing transparency and accountability. [2]
- 4 Retail customer segmentation**
- A large retail chain uses machine learning to segment its customer base to personalize marketing strategies and improve customer service.
- a Discuss whether supervised or unsupervised learning is more appropriate for customer segmentation. [4]
 - b Outline the potential benefits of using cloud computing resources over in-house servers for processing large customer data sets. [2]
 - c Outline the role of outlier detection in customer segmentation. [2]
 - d i Identify a clustering algorithm suitable for handling large data sets with high dimensionality. [1]
 - ii Describe a reason for your choice. [2]
 - e Describe the potential privacy issues that may arise from the detailed segmentation of customers' buying habits. [3]
 - f Describe strategies to mitigate the risk of discriminatory marketing practices that could result from biased data in customer segmentation. [2]
- 5 Natural-disaster prediction and management**
- A government agency deploys machine learning models to predict natural disasters such as floods and earthquakes, aiming to enhance preparedness and response strategies.
- a Describe the application of neural networks in predicting natural disasters and the kind of data they might process. [2]
 - b Describe the impact of using real-time data processing on system requirements and infrastructure. [2]
 - c Describe the challenges associated with integrating and cleaning data from multiple sources, such as satellite imagery and geological sensors. [2]
 - d Describe the role of data augmentation in improving the accuracy of predictions in areas with sparse historical data. [2]
 - e Describe the use of deep learning over traditional models for predicting complex natural disaster patterns. [2]
 - f Describe how machine learning models can be trained to adapt to new types of disaster data over time. [2]
 - g Describe the ethical implications of false positives and false negatives in disaster prediction models. [3]
 - h Describe protocols for data governance that may ensure sensitive geographical and personal data used in predictions are protected. [2]
- 6 Automated cyberbullying detection system**
- A software company is developing an automated system to detect and flag instances of cyberbullying on social-media platforms using natural language processing and machine learning.
- a i Define "natural language processing (NLP)". [1]
 - ii Describe its relevance in detecting cyberbullying. [2]
 - b i Describe the computational challenges associated with processing large volumes of social-media data in real time. [2]
 - ii Outline appropriate hardware solutions. [2]
 - c Describe the potential preprocessing steps needed for textual data from social-media posts to prepare it for machine learning models. [2]
 - d Explain the importance of handling sarcasm and ambiguities in text when setting up preprocessing pipelines for detecting cyberbullying. [2]
 - e Describe the use of analytical rule-based systems vs machine learning models in the context of cyberbullying detection. [2]
 - f Outline the ethical considerations of implementing an automated cyberbullying detection system, particularly regarding false positives and false negatives. [3]
 - g Describe the potential privacy implications of analysing users' social-media content, even for the purpose of detecting cyberbullying. [2]